

A History of Modern Numerical Linear Algebra

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What is Numerical Analysis? (a.k.a Scientific Computing)

The definitions are becoming shorter...

- Webster's New Collegiate Dictionary (1973):
"The study of quantitative approximations to the solutions of mathematical problems including consideration of the errors and bounds to the errors involved."
- The American Heritage Dictionary (1992):
"The study of approximate solutions to mathematical problems, taking into account the extent of possible errors."
- L. N. Trefethen, Oxford University (1992):
"The study of algorithms for the problems of continuous mathematics."

Numerical Analysis: How It All Started

Numerical analysis motivated the development of the earliest computers.

- Ballistics
- Solution of PDE's
- Data Analysis

Early pioneers included:

J. von Neumann, A. M. Turing

In the beginning

von Neumann & Goldstine (1947):

”Numerical Inversion of Matrices of High Order”

Numerical Linear Algebra

Numerical Linear Algebra (NLA) is a small but active area of research: Less than 200 active, committed persons. But the community involves many scientists.

Top Ten Algorithms in Science (Jack Dongarra, 2000)

- 1 Metropolis Algorithm for Monte Carlo
 - 2 Simplex Method for Linear Programming
 - 3 Krylov Subspace Iteration Methods
 - 4 The Decompositional Approach to Matrix Computations
 - 5 The Fortran Optimizing Compiler
 - 6 QR Algorithm for Computing Eigenvalues
 - 7 Quicksort Algorithm for Sorting
 - 8 Fast Fourier Transform
 - 9 Integer Relation Detection
 - 10 Fast Multipole Method
- Red: Algorithms within the exclusive domain of NLA research.
 - Blue: Algorithms strongly though not exclusively) connected to NLA research.